

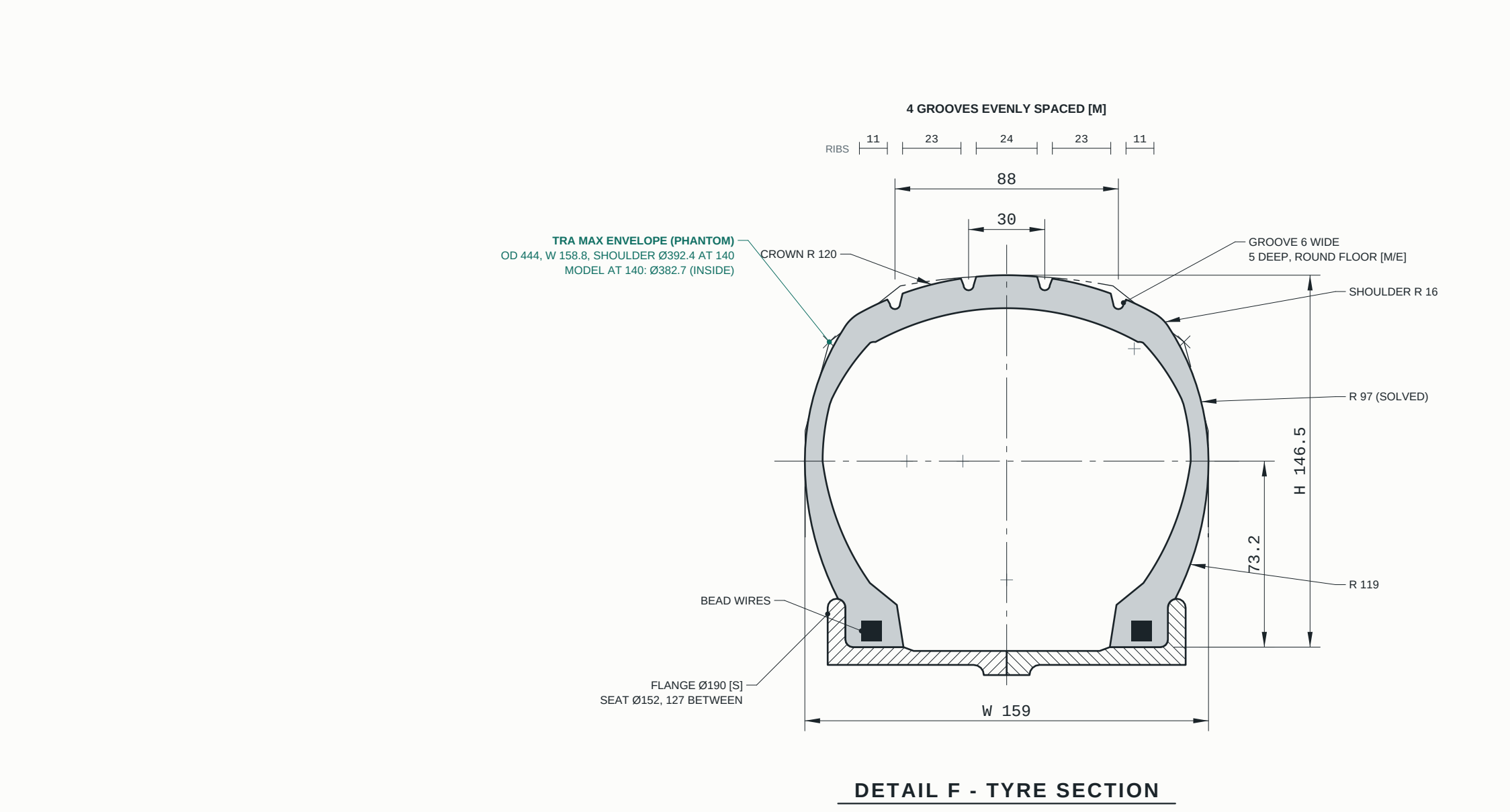
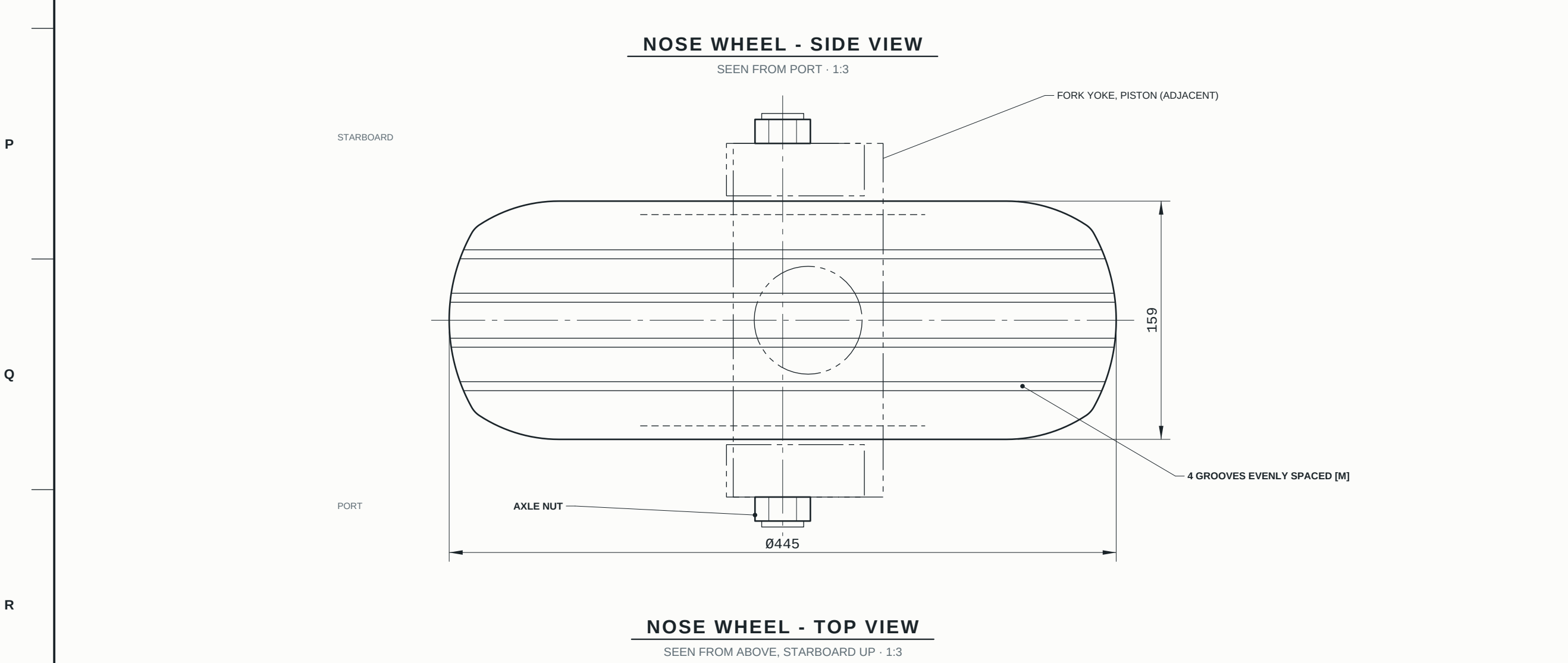
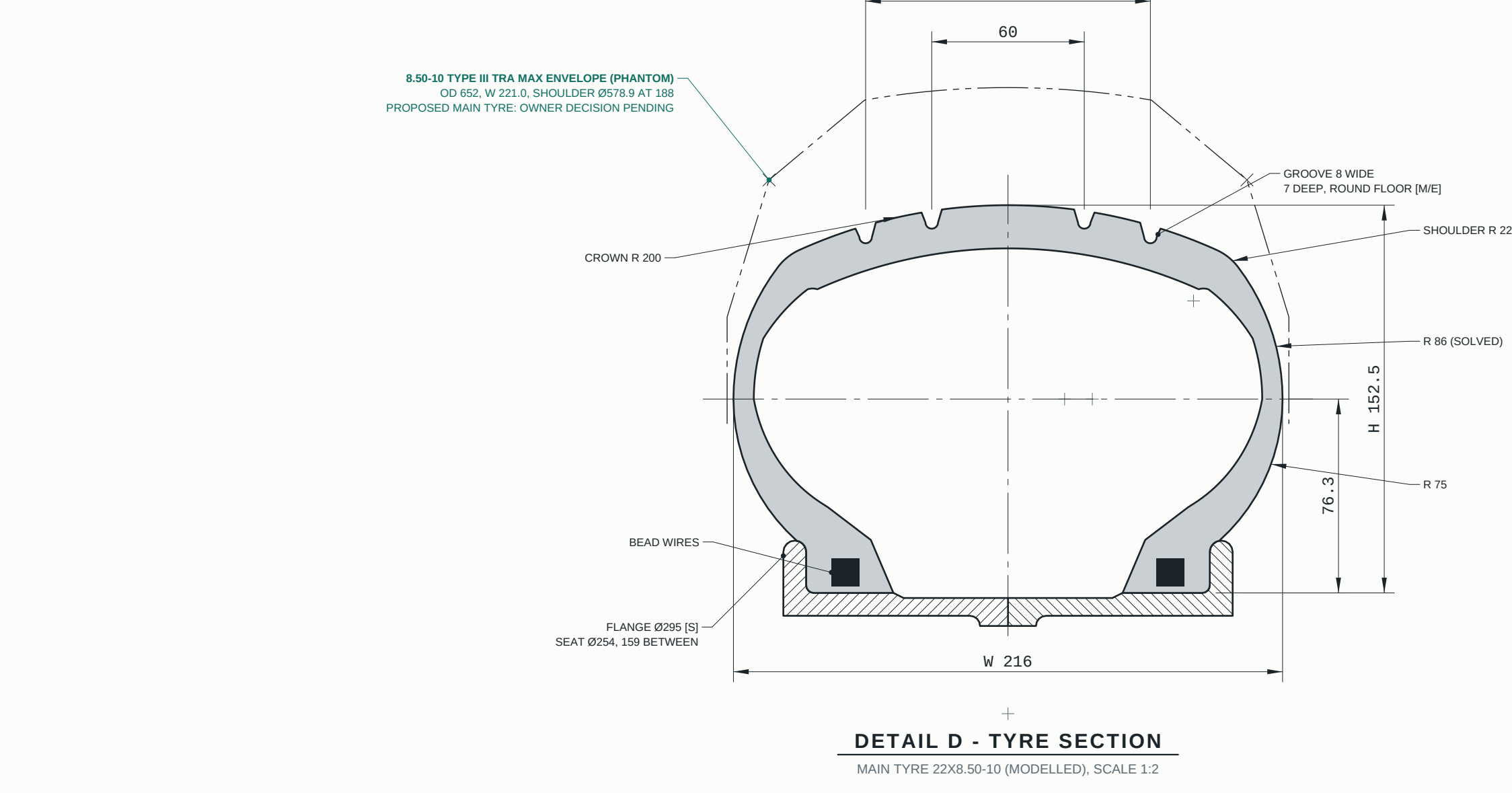
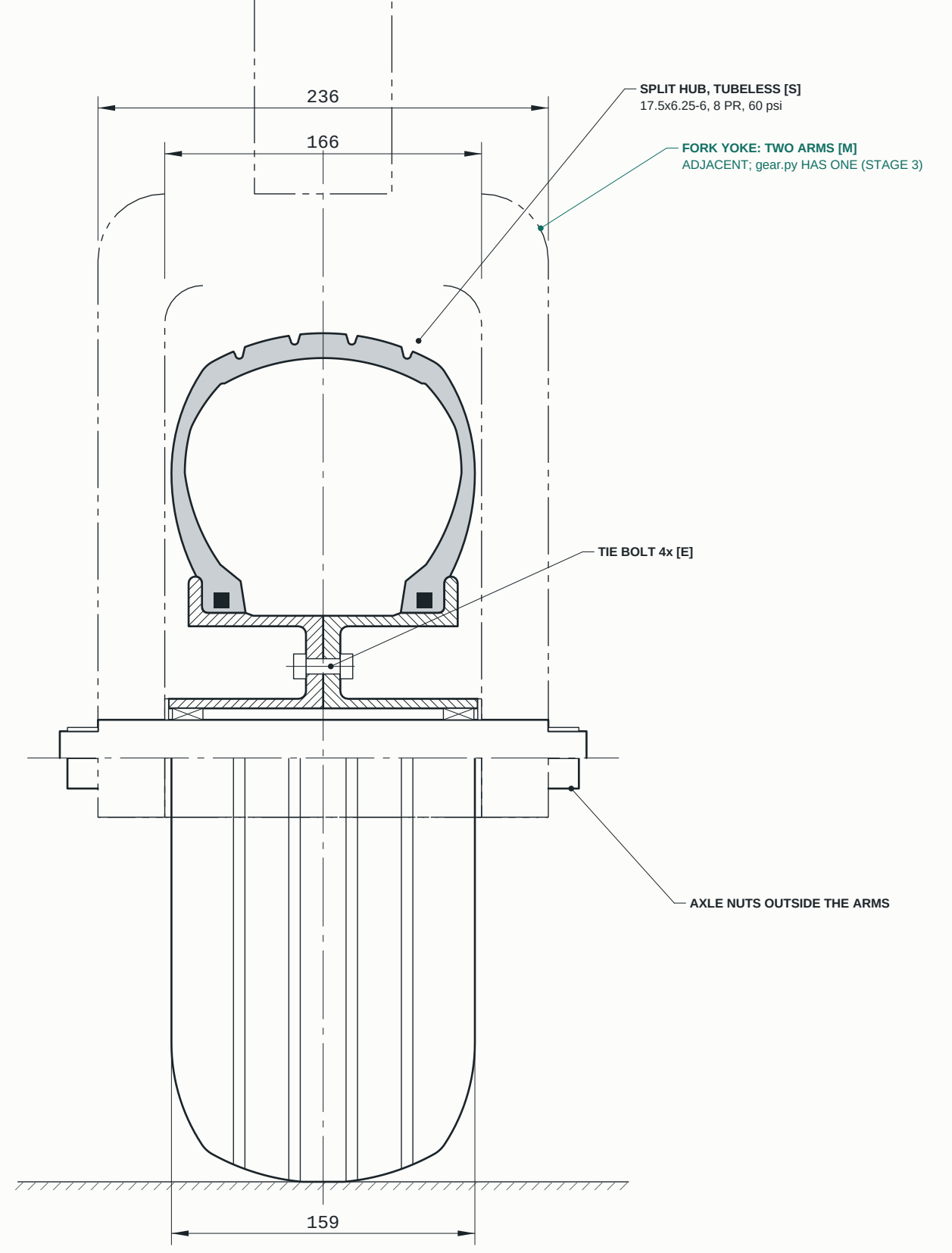
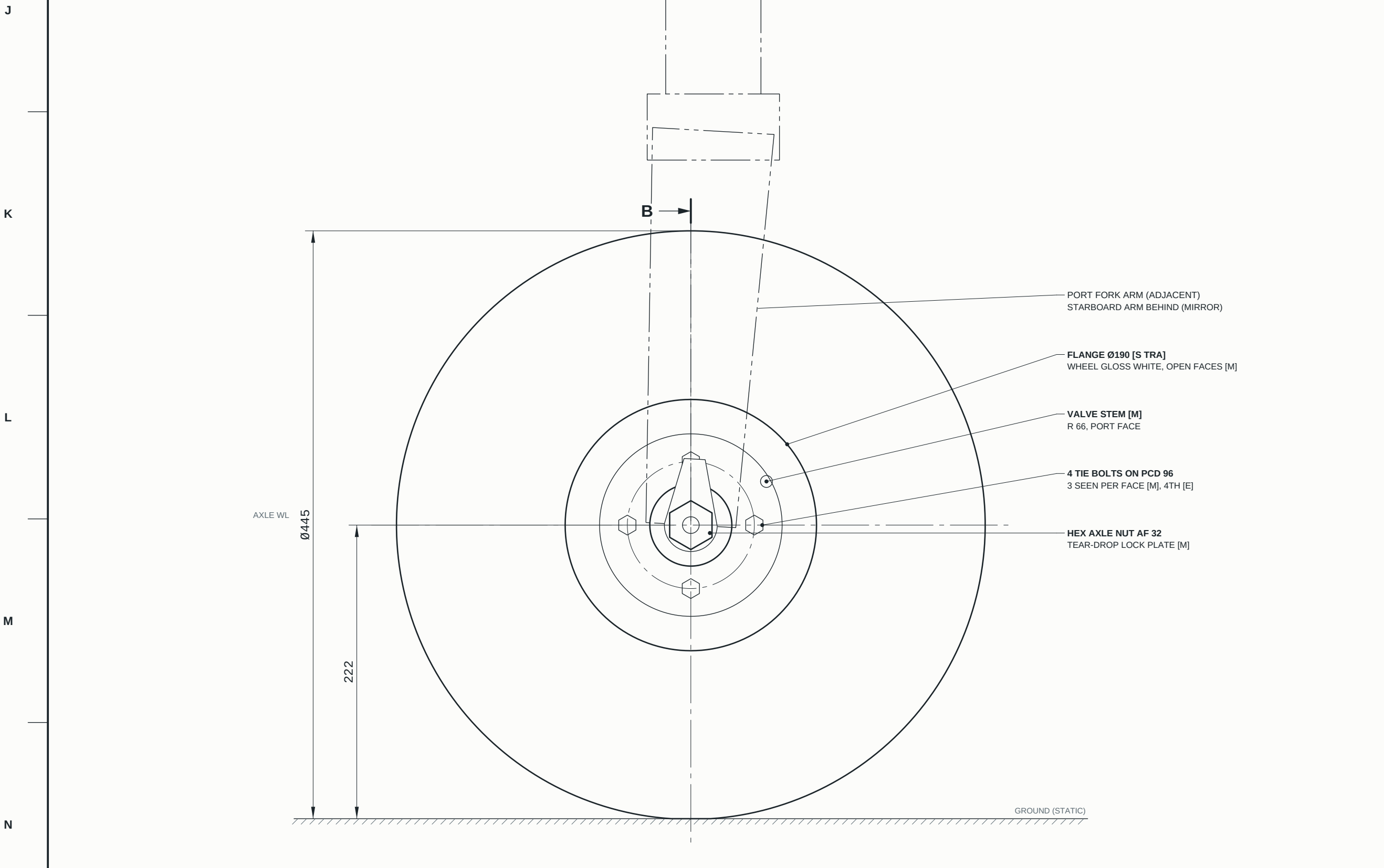
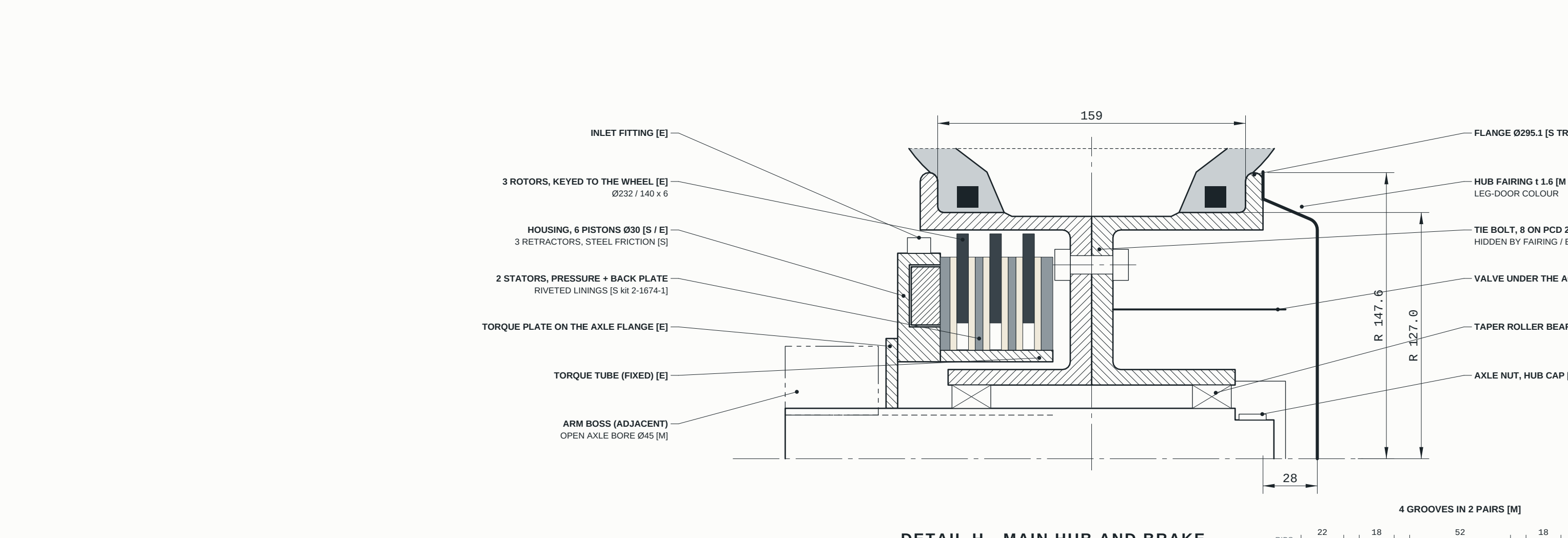
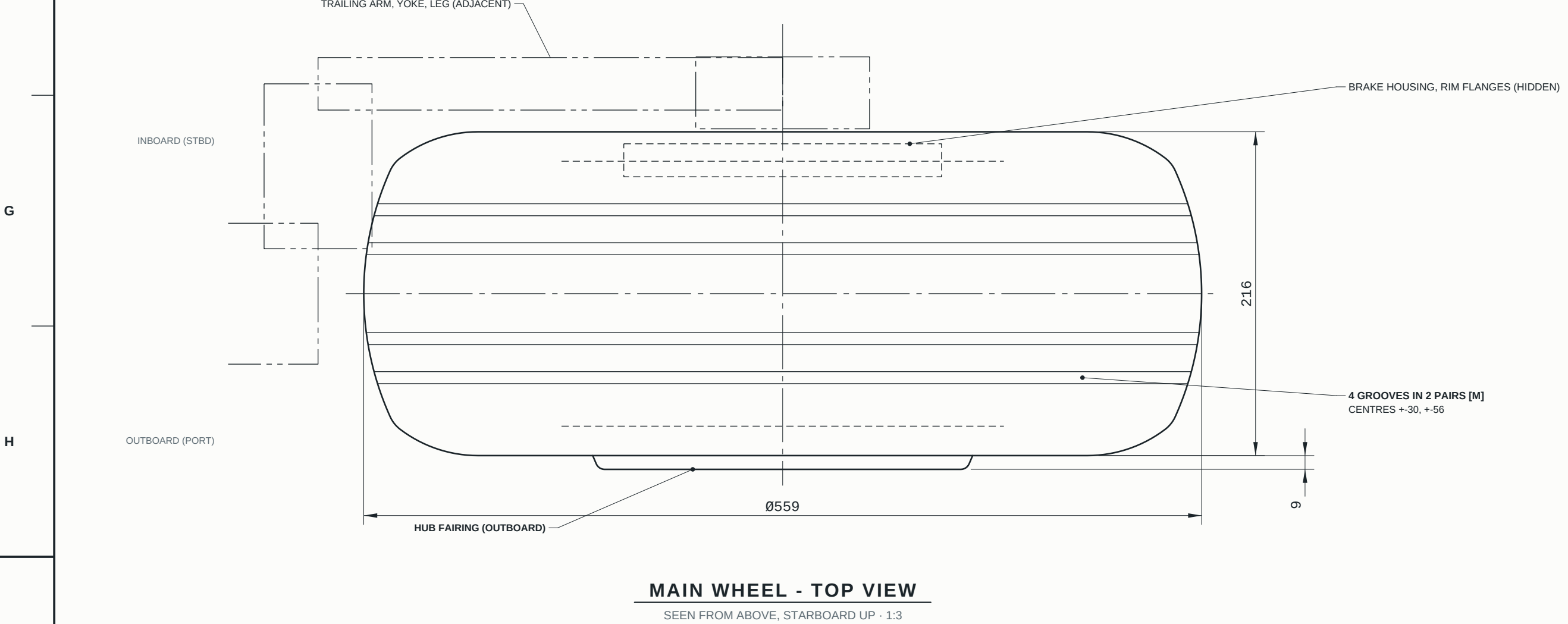
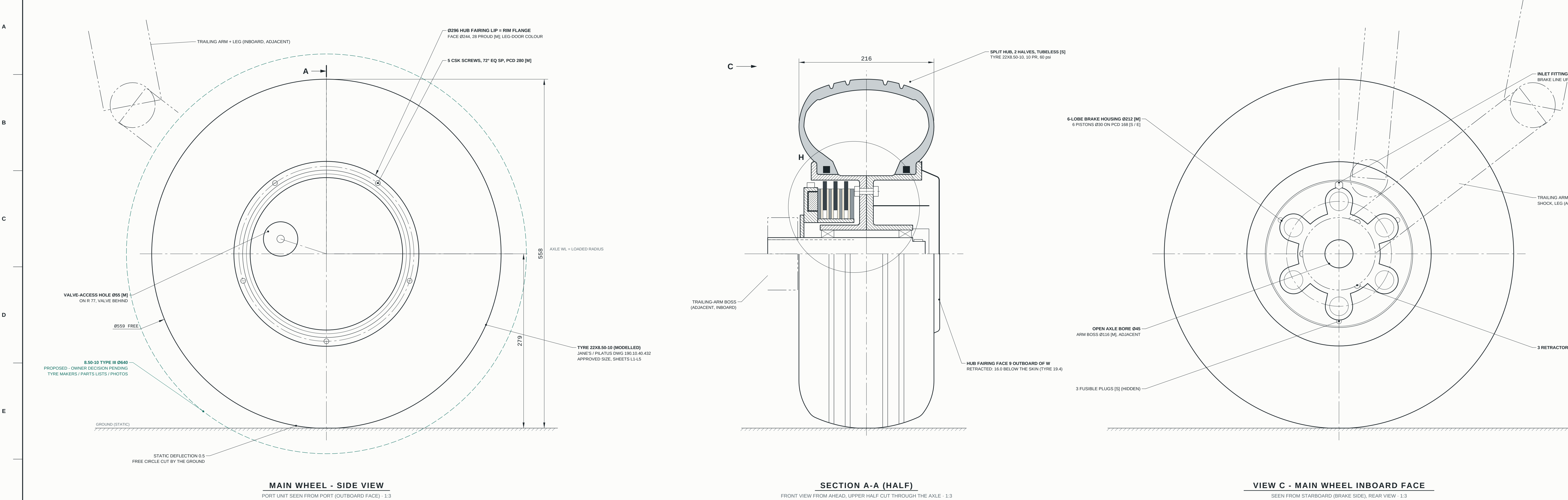
KEY DIMENSIONS - modelwheels.py TABLES				
ITEM (mm)	MAIN	NOSE	SOURCE	
Tyre (modelled)	22x8.58-18	17.5x6.25-6	5 Jane's / Pilatus dwg (approved); P 8.58-18 proposed	
Part numbers	- (not a TRA size)	1706881 / 651-327-6	5 Goodgear / Michelin (parts listings)	
ply / type / pressure	18 PR TL, 60 psi	8 PR TL, 60 psi	5 Oates book, PWE placard, 556-82527	
Free OD / section W	559 / 216	445 / 159	5 22 in / 8.58 in (TRA 8.58-10: 627-652)	
Axle M = loaded R / defl.	279 / 8.5	222 / 6.5	6 axle kept; M main 278-288	
Contact patch / bleed R	38 (chord)	152 / 196	6 Free circle cut by the ground	
Section H / aspect	152.5 / 6.71	146.5 / 6.02	6 (TRA 8.96 / 6.92)	
Head wheel / flange dia	254 / 252.1	152 / 150	5 TRA 158-20 / 6.25; flange .85 / .75 in	
between Flanges / ledge	158.8 / 34.3	127 / 22.9	5 TRA 6.25 / 5.08 in; 1.35 / 0.99 in	
Crown R / tread width	288 / 164	128 / 116	5 main; M nose / M	
Shoulder R / sidewall	12 / 86	18 / 97	5 / 6 (calved tangent)	
grooves: centres +-	36, 56 (2 pairs)	15, 44 (even)	5 head-on photos 3696, 3968, 3981	
Groove width x depth	8 x 7	6 x 5	5 width / E depth	
Main centre / air / outer	52 / 18 / 22	28 / 23 / 11	5 outer rib to the shoulder	
wheel	split hub, 2 halves	split hub, 2 halves	5 POW 7-4-11; Goodrich 3-1543 / 3-1581	
Tie bolts n on PCO	8 on 288	4 on 96	5 hidden (mini); M PCO, E count (nose)	
Valve / safety plug	M 77 / 1	M 66 / 1	5 position; S rim	
flexible plugs	3 (inboard half)	-	5 POW 7-4-11; E position	
Hub flaring lip / face / proud	286 / 244 / 28	-	5 3036, 3966, 3916, 3969, NIX	
Flaring screws / hole	5 on 286 / 465, 6077	-	5 S < 62 deg	
Brake	6 pist., 3 retr., steel	-	5 POW 7-4-10; kit 2-1674-1	
Brake discs	3 rotors / 2 stators	-	5 3 rotors / 2 stators	
Housing lobes / rotor dia	212 / 232	-	5 lobe tips / E	
Brake stack S	189 ... 29	-	5 E (< 8 inboard)	
Axle dia / bore	52 / 49	40	5 gear py / M open bore	
Fork arms (faces)	-	+82...118	5 two arms (head-on photos)	

PHOTO CHECK - DEVIATION CALL-OUTS (mm)				
ITEM	PHOTOS (M)	TABLE / MODEL	A	ACTION
Main tyre OD (modelled)	628-659	559	-81	owner decision pending
8.58-18 Tyre III (proposed)	-	640	8	MAIN_TYRE_CHOICE
3-D gear MAIN_TYRE	-	559	8	From wheels.py
Main section width	214-224	216	0	
Main loaded radius	278-288	axle 279	0	Free circle
Tyre R / flaring lip R	2-15-2-11	2.89	-0.26	
retracted tyre depth	-25 (POW 1 in)	19.4	-6.0	20-28 (fit_check 5)
Main grooves / W	6.14, 8.26	6.14, 8.26	0	3-D geometry
Reorder groove shader	off (geometry)	-	done	
Nose tyre OD	17.5 in	445	0	-
Nose loaded radius	288-218	axle 222	+12.22	axle kept
Nose grooves / W	6.89, 8.28	6.89, 8.28	0	-
Nose fork arms	2	gear.py 1	-1	Stage 3
Hub flaring dia / proud	258-288 / 35-38	286 / 28	0	Face 9 > M/2
Main pressure	60 psi placard	gear.py info 60	0	
Main brake	6-lobe, in wheel	3-D: wheels.py	8	-

MATERIALS (model/assembly.py NAMES)				
TYPE	PART	MAIN	NOSE	NOTE
Tyre	tyre	tyre	tyre	black satin; NO lettering
wheel halves	wheel	wheel	wheel	gloss light grey / white
Hub flaring	hub, gear_dscr	metal / metal_dscr	-	leg-steel colour (1988 blue)
Brake housing / discs	-	-	-	dull alu / dark steel
Bolts, valve stem	-	cadmium	cadmium	yellow-chrome cadmium / brass
Flaring screws, valve cap	-	metal / black	-	rubber-scaled cap
Axle nut, hub cap	steel	steel	steel	lock plate (nose)
Arm / fork	gear_leg	gear_leg	gear_leg	adjacent

LEGEND		SCALE 1:3
	tyre (rubber) in section	
	metal in section (hatched)	
	break/cover/hidden line	
	adjacent parts, free type, TDA arm	
	hidden	
	centre line, pitch circle	
	8.58-18 tyre in type (proposed)	

- NOTES
- Drawn from the parameter tables of modelwheels.py (PTable: value = source tag + note): tyre envelopes MAIN_TYRE_EHV / NOSE_TYRE_ENV, sections, rims, flaring, brake, axle; never from the mesh. Axle positions and the gear context (trailing arm, leg, shock, nose fork) from modelgear.py, drawn as adjacent parts (phantom). Photos and their measurement working stay in the g-ignored namespace.
 - First-angle projection. Each assembly: SIDE view from part (phantom), FRONT view from ahead to its right (upper half in section through the axle), TOP view from above below it; for the main wheel also view C from starboard (inboard / brake face) at the far right. Each group's side / front views stand on one ground line.
 - MAIN_TYRE: modelled 22x8.58-18 (Jane's / Pilatus drawing 190-10-40-422: the size of PC12CALCULATED and its source facts and the approved sheets L14, OD 559). PROPOSED: OWNER DECISION PENDING: the 8.58-18 Tyre II of the free makers, parts list and four photo methods (TRA OD 102-652: crown free OD 440; 22x8.58-10: no inboard tyre bias); shown in the adjacent colour. One line switches 6 (wheels MAIN_TYRE_CHOICE PC12_MAIN_TYRE-8.58-10 for a trial build); axles and static stance kept (axle WL 279 + its loaded radius, flat on the ground over a 250 mm patch, 41 mm deflection).
 - 3-D model gear MAIN_TYRE = this envelope (R 279.5); the LD-1 kg door scallop (R = 12.5 + 202.5) and the round well (lays WELL_R) below it: modelwheels.py main_wheel / nose_wheel involute / extrude the profiles of this shoe tyre with the grooves, wheel halves, hub flaring, brake, valve, its hub(s), subsequently, 2028.py (LAW) compares the mesh with them.
 - Type section: tangent axis - crown R to the read half-width, shoulder round, upper sidewall (radius solved: tangent to the shoulder, max width W02 at 0.50 H), lower sidewall to where the tyre leaves the rim-flange lip round. The 22 in size has no TRA entry; detail D shows the proposed 8.58-18 TRA envelope.
 - Main hub being (PCH) flaring on the outer half(s), paired in the leg door contour, its face stands 9 mm outboard of the tyre's max width, retracted (when fitted with the skin) 18 mm below the skin, inside the tyres 19.4 mm (POW-1 in, fit_check 5).
 - Main brake: six pistons, three retractors, steel friction surfaces, bolted to the axle (POW 7-4-10); 3 rotors / 2 stators inferred from the overhaul kit 04 pads, 2 stators, 48 rotors). Housing, stack and torque tube sized inside the robust wheel half (E): one piston side is shown in section.
 - No markings (owner decision): no sidewall lettering, brand names or placards on wheels or tyres. The tread grooves are geometry (the render-time groove shader is off). Screw, valve and tie-bolt locking is the drawn pose (the wheels turn): starboard units are mirror images. The main leg door (LD-1) is not shown.



PILATUS PC-12 PRO

WHEELS & TYRES - MAIN AND NOSE WHEEL ASSEMBLIES

OWG NO

REV

SCALE

SHEET

PC12-L4W

A

1:3 / 1:2

A0 - 1 OF 1

UNITS

DATE

mm

2026-09-24

DRAWN

Claude - loftkit parametric model

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION